

ORIGINAL RESEARCH

Transcatheter Aortic Valve Replacement or Medical Treatment for Paradoxical Low-Flow, Low-Gradient Aortic Valve Stenosis

REBOOT-PARADOX

Julinda Mehilli, MD,^{a,b,*} Julius Steffen, MD,^{b,c,*} Tanja Rudolph, MD,^d Magda Haum, MD,^{b,c} Christina Giannini, MD, PhD,^e Konstantin Stark, MD,^{b,c} Hüseyin Ince, MD,^f Lenard Conradi, MD,^{g,h} Michael Joner, MD,^{c,i} Andreas Schäfer, MD,^g Ann-Katrin Karcher, MD,^j Martin Landt, MD,^k Matti Adam, MD,^l Holger Thiele, MD,^m Tobias Geisler, MD,ⁿ David Jochheim, MD,^b Christian Hagl, MD,^{c,o} Uwe Zeymer, MD,^{j,p} Stephan Windecker, MD,^q Steffen Massberg, MD,^{b,c} on behalf of the REBOOT-PARADOX-Trial Investigators

ABSTRACT

BACKGROUND Paradoxical low-flow, low-gradient (pLFLG) aortic stenosis (AS) has an adverse prognosis in comparison to other AS subtypes. Transcatheter aortic valve replacement (TAVR) has never been compared with optimal medical therapy (OMT) alone in a dedicated randomized trial in this population.

OBJECTIVES The trial sought to determine whether TAVR added to OMT reduces all-cause mortality in symptomatic patients with pLFLG AS.

METHODS In this multicenter, open-label trial, patients with symptomatic pLFLG AS (aortic valve area ≤ 1.0 cm², mean gradient < 40 mm Hg, stroke volume index < 35 mL/m², ejection fraction $\geq 50\%$) were randomly assigned 2:1 to TAVR plus OMT or OMT alone. The primary endpoint was all-cause death, assessed 2 years after enrollment of the last patient. Enrollment was stopped prematurely for slow recruitment.

RESULTS Of 783 planned patients, 120 (median age 82 years; 53.3% women) were randomized (80 TAVR, 40 OMT). A total of 19 OMT patients eventually crossed over to TAVR, most for symptom progression. The primary endpoint occurred in 29.0% vs 37.1% (HR: 0.83; 95% CI: 0.41-1.66; $P = 0.60$), and 5-year mortality was 44.9% vs 56.6% (HR: 0.76; 95% CI: 0.43-1.35; $P = 0.35$). TAVR was associated with improved symptoms (NYHA functional class I/II at 1 year, 83.0% vs 59.0%) and lower cumulative incidence of first endpoint-related rehospitalizations, and there were no differences in functional capacity or quality of life.

CONCLUSIONS Although the effect of TAVR on mortality remains unresolved, these findings support an individualized strategy with close surveillance and timely TAVR upon clinical deterioration in selected patients with pLFLG AS. Adequately powered randomized trials are needed to define the effect of early TAVR on prognosis. (accuRate Evaluation of Benefit with Optimal medical treatment with or withOut Transcatheter valve repair of PARADOXical low flow low gradient aortic stenosis [REBOOT-PARADOX Trial]; [NCT03863132](#)) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS**AS** = aortic valve stenosis**AVA** = aortic valve orifice area**AVR** = aortic valve
replacement**dPmean** = mean transvalvular
gradient**HFpEF** = heart failure with
preserved ejection fraction**LVEF** = left ventricular
ejection fraction**OMT** = optimal medical
therapy**pLFLG** = paradoxical low-flow
low-gradient aortic stenosis**SVI** = stroke volume index**TAVR** = transcatheter aortic
valve replacement

Transcatheter aortic valve replacement (TAVR) is established as the preferred treatment for many patients with symptomatic aortic valve stenosis,¹ supported by randomized trials demonstrating superior outcomes compared with medical therapy and noninferiority or superiority to surgical aortic valve replacement across a broad range of surgical risk categories.¹⁻⁴

Most patients with severe AS present with a high transvalvular pressure gradient (mean gradient ≥ 40 mm Hg). However, approximately 20% of patients with AS have paradoxical low-flow, low-gradient (pLFLG) AS, characterized by a mean transvalvular gradient < 40 mm Hg despite preserved left ventricular ejection fraction (LVEF), and a stroke volume index (SVI) < 35 mL/m².^{1,5} A meta-analysis of 18 studies including 7,459 patients demonstrated significantly higher mortality in pLFLG AS than in high-gradient AS, underscoring the adverse prognosis of this phenotype.⁶ This excess risk likely reflects the complex clinical profile of pLFLG AS, which is frequently accompanied by concomitant cardiac conditions such as mitral or tricuspid regurgitation, atrial fibrillation, and diastolic dysfunction resembling heart failure with preserved ejection fraction (HFpEF).⁷ Although AVR, including TAVR, is recommended for symptomatic patients with pLFLG AS, the supporting evidence is limited.¹ To date, it is derived primarily from subgroup analyses of randomized trials⁸ that largely enrolled patients with high-gradient AS, together with retrospective registry studies.⁹ These data

suggest that patients with pLFLG AS experience less symptomatic improvement and smaller survival benefits after AVR than patients with high-gradient AS.^{10,11} However, whether TAVR confers a survival benefit over optimal medical therapy in patients with pLFLG AS has never been evaluated in a dedicated randomized clinical trial.

The REBOOT-PARADOX (accuRate Evaluation of Benefit with Optimal medical treatment with or without Transcatheter valve repair of PARADOXical low flow low gradient aortic stenosis; [NCT03863132](#)) trial was designed to determine the effect of TAVR in addition to optimal medical therapy (OMT) on mortality in symptomatic patients with pLFLG AS. Premature termination limited the precision of the mortality estimate. However, the randomized design, detailed long-term patient-level follow-up, and frequent crossover to TAVR provide a unique opportunity to characterize the clinical course with upfront TAVR vs an initially conservative strategy with OMT and subsequent TAVR when clinically indicated in this heterogeneous population.

METHODS

STUDY DESIGN AND OVERSIGHT. Patients were enrolled at 15 centers in Germany, Italy, and Switzerland ([Supplemental Table 1](#)). The trial was designed by the steering committee. IHF GmbH - Institut für Herzinfarktforschung (Ludwigshafen, Germany) served as the trial sponsor and was responsible for trial coordination and data management. An independent data safety monitoring board (DSMB) was going to review safety data in a meeting after recruitment of 150 patients. This was not carried

From the ^aLA-Regio Kliniken, Klinik Landshut-Achdorf, Landshut, Germany; ^bMedizinische Klinik und Poliklinik I, LMU University Hospital, LMU Medizin, Ludwig-Maximilians-Universität München, Munich, Germany; ^cDZHK (German Centre for Cardiovascular Research), Munich Heart Association (MHA), Munich, Germany; ^dKlinik für Kardiologie, Herz- und Diabeteszentrum Nordrhein-Westfalen Bad Oeynhausen, Bad Oeynhausen, Germany; ^eAzienda Ospedaliero Universitaria Pisana, Pisa, Italy; ^fKlinik und Poliklinik für Kardiologie, Department für Innere Medizin, Universitätsmedizin Rostock, Rostock, Germany; ^gUniversitäres Herz- und Gefäßzentrum Hamburg, Klinik und Poliklinik für Herz- und Gefäßchirurgie, Universitätsklinikum Hamburg-Eppendorf, Hamburg, Germany; ^hHerzzentrum der Universität Köln, Klinik und Poliklinik für Herzchirurgie, Uniklinik Köln, Cologne, Germany; ⁱTUM Universitätsklinikum Deutsches Herzzentrum, Technische Universität München, Munich, Germany; ^jMedizinische Klinik B, Klinikum der Stadt Ludwigshafen am Rhein gGmbH, Ludwigshafen, Germany; ^kHerzzentrum, Kardiologie und Angiologie, Segeberger Kliniken, Bad Segeberg, Germany; ^lDepartment of Cardiology, Heart Center Cologne, University of Cologne, Faculty of Medicine and University Hospital, Cologne, Germany; ^mDepartment of Cardiology and Leipzig Heart Science, Heart Center Leipzig at Leipzig University, Leipzig, Germany; ⁿInnere Medizin III, Universitätsklinikum Tübingen, Tübingen, Germany; ^oHerzchirurgische Klinik und Poliklinik, LMU University Hospital, LMU Medizin, Ludwig-Maximilians-Universität München, Munich, Germany; ^pDepartment of Cardiology and Angiology, University Heart Center Freiburg-Bad Krozingen, Faculty of Medicine, University of Freiburg, Freiburg, Germany; and the ^qDepartment of Cardiology, Inselspital, Bern University Hospital, University of Bern, Bern, Switzerland. *These authors contributed equally to this work.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

out due to early termination of patient enrollment. An independent event adjudication committee (EAC), whose members were unaware of treatment assignments, adjudicated all deaths and cardiovascular serious adverse events, including rehospitalizations. Echocardiographic studies were analyzed by an independent core laboratory (Yale Echo Core Lab, Yale University School of Medicine, New Haven, Connecticut, USA).

The trial was conducted in accordance with the Declaration of Helsinki, Good Clinical Practice, ISO 14155, and applicable national regulations. The protocol was approved by the ethics committee at each participating site, and all patients provided written informed consent before enrollment. The trial was registered before enrollment of the first patient ([NCT03863132](#)). Enrollment began in December 2019. Owing to slower-than-anticipated recruitment, potentially also affected by the COVID-19 pandemic, enrollment was terminated prematurely on September 1, 2022, by the steering committee in agreement with the funding sponsor after 120 patients had undergone randomization. Data-lock date for the primary analysis was July 8, 2026.

PATIENT POPULATION. Patients aged ≥ 18 years with symptomatic native aortic valve stenosis were eligible if they had an aortic valve orifice area (AVA) of ≤ 1.0 cm² (or indexed AVA ≤ 0.6 cm²/m²), a mean transvalvular gradient (dPmean) of < 40 mm Hg, an SVi of < 35 mL/m², and an LVEF $\geq 50\%$. In addition, the diagnosis of severe AS had to be confirmed by either multidetector computed tomography demonstrating severe aortic valve calcification or fulfillment of prespecified echocardiographic likelihood criteria. Eligibility for TAVR required consensus by the local heart team. Key exclusion criteria included hemodynamic instability or cardiogenic shock, a preexisting prosthetic heart valve in any position, concomitant severe valvular heart disease, and congenital unicuspid or bicuspid aortic valve. The complete inclusion and exclusion criteria are listed in [Supplemental Table 2](#).

RANDOMIZATION. Eligible patients were randomly assigned in a 2:1 ratio to undergo TAVR plus OMT or to receive OMT alone. Randomization was performed centrally in permuted blocks of randomly varying size, stratified by site. Randomization took place during the index hospitalization on the day written informed consent was obtained. Owing to the nature of the intervention, patients and treating physicians were not blinded to treatment assignment; the EAC remained blinded, and observer blinding (eg, laboratory personnel) was maintained wherever possible.

STUDY PROCEDURES. In patients assigned to TAVR, the procedure was to be performed within 7 days after randomization, preferably via transfemoral access in accordance with contemporary guideline recommendations and institutional practice. Implantation of the self-expanding ACURATE neo valve (Boston Scientific) was recommended but not mandated. A balloon-expandable transcatheter heart valve could be used at the discretion of the local heart team if anatomical considerations precluded implantation of the self-expanding device. Periprocedural anticoagulation consisted of unfractionated heparin (50-70 IU/kg of body weight), adapted to individual patient needs, and the post-TAVR antithrombotic regimen followed local institutional protocols. Patients assigned to the control group received OMT according to current guidelines¹² and institutional standards.

Crossover from the OMT group to TAVR was permitted for predefined clinical indications, including new-onset left ventricular systolic dysfunction (LVEF $< 50\%$) not attributable to another cause, an increase in the mean transvalvular pressure gradient of > 20 mm Hg within 6 months, recurrent hospitalizations for heart failure attributable to AS, or progressive symptoms resulting in an unacceptable quality of life despite OMT.

FOLLOW-UP. Patients underwent follow-up at 30 days, 12 months, 24 months, and 5 years after randomization. Follow-up at 30 days and 5 years was performed by telephone, whereas 12- and 24-month assessments were conducted during in-person study visits that included structured evaluation of cardiac function and symptoms.

ENDPOINTS. The primary endpoint was the time from randomization to death from any cause 2 years after enrollment of the last patient. All deaths were adjudicated by an independent EAC whose members were unaware of treatment assignments and who applied prespecified endpoint definitions.

Secondary endpoints included periprocedural and cardiovascular mortality, changes in echocardiographic parameters, functional status (NYHA functional class), neurological events (stroke or transient ischemic attack), cardiac conduction disturbances, major adverse cardiovascular and cerebrovascular events, and quality of life. Prespecified TAVR-specific procedural endpoints were assessed only among patients assigned to the TAVR group. A detailed list of all endpoints is provided in [Supplemental Table 3](#).

Rehospitalizations were analyzed as the cumulative incidence of the first unplanned rehospitalization related to AS, heart failure, or prosthetic valve

complications. Heart-failure rehospitalizations were analyzed separately as a predefined subgroup of these events. Hospitalizations solely for elective crossover TAVR were not considered endpoint rehospitalizations. However, if crossover TAVR was performed during a hospitalization for worsening heart failure or symptomatic deterioration, the hospitalization itself was counted as the endpoint event. All rehospitalization events were prospectively collected in both treatment groups and centrally adjudicated by the independent EAC.

BLINDING. Although REBOOT-PARADOX was open-label, blinding procedures were prespecified in a blinding manual. The primary endpoint was analyzed by a statistician blinded to treatment allocation, using analysis programs developed on a reduced, treatment-masked dataset and cross-validated by independent programming. The EAC and Steering Committee were blinded to group assignment as detailed therein. For the data safety monitoring board meeting, blinding had also been planned.

STATISTICAL ANALYSIS. The trial was originally designed to test the superiority of TAVR plus OMT over OMT alone with respect to all-cause mortality using a 1-sided log-rank test at a significance level of 0.025. The sample size calculation assumed exponential survival, with a 2-year mortality of 50% in the OMT group and 40% in the TAVR group, linear enrollment over 24 months, a crossover rate from OMT to TAVR of 11.7% during the first year, and a dropout rate of 2.2%. Under these assumptions, randomization of 783 patients in a 2:1 ratio (522 assigned to TAVR and 261 to OMT) was estimated to provide 80% power for the primary analysis, planned 2 years after enrollment of the last patient.

Enrollment was terminated prematurely after 120 patients had undergone randomization. Consequently, the trial did not achieve the planned sample size, and all efficacy analyses should be regarded as exploratory and hypothesis-generating rather than confirmatory.

The primary analysis was performed according to the intention-to-treat principle in the full analysis set, which included all randomized patients analyzed according to their assigned treatment group, irrespective of the treatment received. A supportive treated-as-randomized analysis includes patients who received their assigned treatment during the index hospitalization. Because of the limited sample size and the high crossover rate following premature trial termination, the prespecified per-protocol analysis was not performed. An additional supportive time-dependent analysis was performed in the

treatment-exposure set, which compares the 22 patients who remained on OMT throughout follow-up with the 98 patients who underwent TAVR at any time during the study period.

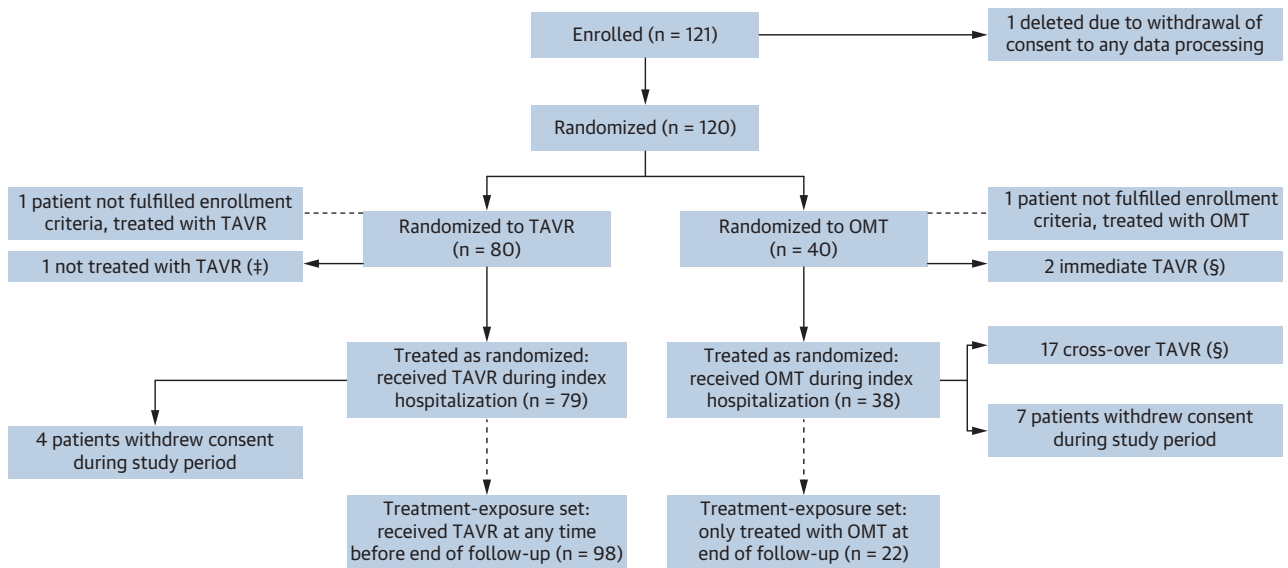
Owing to premature trial termination, the originally planned confirmatory hypothesis test was not performed. Proportional hazards were evaluated using log-log survival plots within the prespecified analysis window. The primary endpoint was analyzed descriptively using a 2-sided log-rank test. Kaplan-Meier estimates were used to describe cumulative event rates. Secondary endpoint analyses were exploratory. The cumulative incidence of first rehospitalization was calculated using the Aalen-Johansen estimator with death as a competing event. Because planned admissions for crossover TAVR were excluded by definition, the estimates reflect unplanned clinical deterioration rather than procedural scheduling. No adjustment for multiple comparisons was performed.

A time-dependent Cox model was used as an exploratory analysis to account for the fact that TAVR occurred at different times during follow-up. This approach appropriately assigns preprocedure time to the non-TAVR state. However, because the timing of TAVR was not randomized and likely reflects symptoms, physician decision-making, and survival until the procedure, this model cannot restore the benefits of randomization. Therefore, the results should be interpreted as exploratory associations between TAVR exposure and subsequent mortality rather than causal treatment effects.

Continuous variables are presented as mean $\pm$ SD or median (IQR), as appropriate. Categorical variables are presented as counts and percentages. Details of all analyses were prespecified in the statistical analysis plan finalized before completion of patient enrollment. All analyses were performed with SAS software, version 9.4 (SAS Institute Inc).

RESULTS

PATIENTS. Between December 2019 and September 2022, when enrollment was stopped prematurely because of slower-than-anticipated recruitment, 121 patients were enrolled at 15 sites ([Supplemental Table 1](#)); 1 patient withdrew consent to data processing, leaving 120 patients who underwent randomization: 80 were assigned to TAVR plus OMT and 40 to OMT alone ([Figure 1](#)). Among patients assigned to TAVR, 79 of 80 (98.8%) underwent the procedure. Crossover from the OMT group to TAVR was common: 19 of 40 patients (47.5%) underwent TAVR, including 2 shortly after randomization and 17

FIGURE 1 CONSORT Diagram

Study flow diagram showing the flow of the participating patients at each stage of the trial. Boxes on the left and right list protocol deviations and treatment crossovers. Dashed lines show patients not having fulfilled echocardiographic eligibility criteria but who remained part of the analyzed populations. Dashed arrows indicate the regrouping of patients into the treatment-exposure set, which includes only those who received either TAVR or OMT alone at the end of follow-up. §Patient counted in the treatment-exposure set "Only treated with OMT at end of follow-up"; §Patient counted in the treatment-exposure set "Received TAVR at any time before end of follow-up." OMT = optimal medical therapy; TAVR = transcatheter aortic valve replacement.

after meeting prespecified crossover criteria (see below). Consequently, by the end of follow-up, 98 patients had undergone TAVR and 22 had received OMT alone.

The median duration of follow-up was 50.4 months in the TAVR group and 40.9 months in the OMT group. Follow-up was 96.2% complete in the TAVR group and 92.5% complete in the OMT group at the time of the primary analysis; completeness by group and time point, together with deaths and withdrawals, is summarized in [Supplemental Table 4](#). Eleven patients (4 in the TAVR group and 7 in the OMT group) withdrew consent during follow-up.

BASELINE CHARACTERISTICS. Baseline demographic, clinical, and echocardiographic characteristics were generally well balanced between the 2 groups ([Table 1](#)). The median age was 82 years (IQR: 78-85 years), 53.3% of patients were women, and the median EuroSCORE II and STS-PROM (Society of Thoracic Surgeons predicted risk of mortality score) were both 3.0%. Coronary artery disease (57.5%), atrial fibrillation (58.3%), arterial hypertension (88.3%), and chronic kidney disease (27.5%) were common, and 55.8% of patients were in NYHA functional class III or IV. Hypertension was more frequent

in the TAVR group (93.8% vs 77.5%), whereas a history of stroke or transient ischemic attack was more common in the OMT group (15.0% vs 3.8%); otherwise, baseline characteristics were well balanced.

As specified by the eligibility criteria, all patients had pLFLG AS with preserved LVEF: the median AVA was 0.8 cm² (IQR: 0.7-0.9 cm²), the mean transvalvular gradient was 21.3 mm Hg (IQR: 19.0-28.6 mm Hg), the SVi was 29.6 mL/m² (IQR: 25.5-33.0 mL/m²), and the LVEF was 58% (IQR: 55%-60%) ([Table 2](#)). Two patients did not fulfill the echocardiographic eligibility criteria, 1 per each group. For both patients, the violation criterion was SVi >35 mL/m². Guideline-directed medical therapy at baseline was similar in the 2 groups ([Supplemental Table 5](#)).

PROCEDURAL CHARACTERISTICS. Among the 79 patients who underwent TAVR as randomized, the procedure was performed via transfemoral access in 98.7% and under conscious sedation in 86.1% ([Table 3](#)). A self-expanding valve was used in 70.9%, and a balloon-expandable valve was used in 29.1% of patients ([Supplemental Table 6](#)). Technical success according to Valve Academic Research Consortium 3 (VARC-3) was achieved in 97.5% of patients. After TAVR, the median mean prosthesis gradient was

TABLE 1 Baseline Characteristics

	All (N = 120)	TAVR (n = 80)	OMT (n = 40)	P Value
Age, y	82 (78-85)	82 (79-85)	81 (78-85)	0.37
Male	46.7 (56/120)	45.0 (36/80)	50.0 (20/40)	0.70
BMI, kg/m ²	25.9 (23.4-29.6)	25.8 (23.4-29.6)	26.0 (22.6-29.4)	0.59
BSA, m ²	1.8 (1.7-2.0)	1.8 (1.7-2.0)	1.9 (1.7-2.0)	0.68
STS-PROM, %	3.0 (2.0-5.0)	3.0 (2.2-4.9)	3.0 (1.9-5.0)	0.48
EuroSCORE II, %	3.0 (2.0-6.1)	3.0 (2.0-6.4)	4.4 (1.8-5.7)	0.97
Coronary artery disease	57.5 (69/120)	55.0 (44/80)	62.5 (25/40)	0.56
History of myocardial infarction	4.2 (5/120)	2.5 (2/80)	7.5 (3/40)	0.33
History of CABG	10.0 (12/120)	8.8 (7/80)	12.5 (5/40)	0.53
History of stroke/TIA	7.5 (9/120)	3.8 (3/80)	15.0 (6/40)	0.06
Peripheral artery disease	10.0 (12/120)	10.0 (8/80)	10.0 (4/40)	1.00
Pulmonary hypertension	8.3 (10/120)	7.5 (6/80)	10.0 (4/40)	0.73
COPD	10.0 (12/120)	7.5 (6/80)	15.0 (6/40)	0.21
Atrial fibrillation	58.3 (70/120)	58.8 (47/80)	57.5 (23/40)	1.00
Arterial hypertension	88.3 (106/120)	93.8 (75/80)	77.5 (31/40)	<0.05
Diabetes mellitus	35.0 (42/120)	35.0 (28/80)	35.0 (14/40)	1.00
Chronic kidney disease	27.5 (33/120)	31.3 (25/80)	20.0 (8/40)	0.28
Symptoms				
Angina pectoris	25.0 (30/120)	26.3 (21/80)	22.5 (9/40)	0.82
Syncope	11.7 (14/120)	12.5 (10/80)	10.0 (4/40)	0.77
Dizziness	34.2 (41/120)	36.3 (29/80)	30.0 (12/40)	0.55
Leg edema	24.2 (29/120)	27.5 (22/80)	17.5 (7/40)	0.26
NYHA class III/IV	55.8 (67/120)	55.0 (44/80)	57.5 (23/40)	0.85
6-min walk test, m	288.5 (185.0-354.0)	285.0 (191.0-354.5)	291.5 (165.0-349.5)	0.74
Laboratory tests				
NT-proBNP, pg/mL	1110 (443-1865)	1036 (436-1841)	1207 (443-1893)	0.92
Creatinine, mg/dL	1.0 (0.9-1.2)	1.0 (0.8-1.2)	1.0 (0.9-1.1)	0.32
hs-Troponin T, pg/mL	17.5 (14.0-27.0)	18.0 (13.6-28.0)	17.0 (14.0-25.0)	0.86
Hemoglobin, g/dL	12.7 (11.8-13.8)	12.8 (12.0-13.8)	12.6 (11.6-13.8)	0.29

Median (IQR) or % (n/N).

BMI = body-mass index; BSA = body surface area; CABG = coronary artery bypass graft; COPD = chronic obstructive pulmonary disease; hs = high sensitivity; NT-proBNP = N-terminal pro-B-type natriuretic peptide; OMT = optimal medical therapy; STS-PROM = Society of Thoracic Surgeons predicted risk of mortality score; TAVR = transcatheter aortic valve replacement TIA = transitory ischemic attack.

TABLE 2 Imaging

	All (N = 120)	TAVR (n = 80)	OMT (n = 40)	P Value
Echocardiography				
AVA, cm ²	0.8 (0.7-0.9)	0.8 (0.7-0.9)	0.8 (0.7-1.0)	0.20
AVA index, cm ² /m ²	0.4 (0.4-0.5)	0.4 (0.4-0.5)	0.4 (0.4-0.5)	0.65
SVi, mL/m ²	29.6 (25.5-33.0)	30.3 (27.3-33.0)	27.7 (23.3-32.1)	<0.05
LVEF, %	58 (55-60)	59 (55-60)	57 (54-61)	0.81
dPmax, mm Hg	37.0 (31.0-46.0)	36.0 (31.0-49.0)	38.0 (29.8-43.9)	0.51
dPmean, mm Hg	21.3 (19.0-28.6)	21.8 (18.3-29.8)	21.0 (20.0-28.0)	0.42
MR > grade 1	26.6 (25/94)	28.6 (18/63)	22.6 (7/31)	0.62
TR > grade 1	34.5 (30/87)	36.2 (21/58)	31.0 (9/29)	0.81
Computed tomography				
Annulus area, mm ²	448.0 (397.0-512.0)	439.5 (399.6-497.0)	463.0 (392.6-524.7)	0.57
Calcification index, AU/m ²	1040.3 (698.0-1368.0)	1014.2 (539.3-1341.0)	1092.9 (713.1-1543.8)	0.23

Values are median (IQR) or % (n/N).

AR = aortic regurgitation; AVA = aortic valve orifice area; dPmax = maximum transvalvular gradient; dPmean = mean transvalvular gradient; LVEF = left ventricular ejection fraction; MR = mitral regurgitation; SVi = stroke volume index; TR = tricuspid regurgitation.

TABLE 3 TAVR Procedural Characteristics

	TAVR-Treated (n = 79)
Duration of procedure, min	54 (41-73)
Conscious sedation	86.1 (68/79)
General anesthesia	13.9 (11/79)
Transfemoral access	98.7 (78/79)
Transapical access	1.3 (1/79)
Predilatation	55.7 (44/79)
Postdilatation	17.7 (14/79)
Cerebral embolic protection	0.0 (0/79)
Type of valve	
Self-expandable	70.9 (56/79)
Balloon-expandable	29.1 (23/79)

Values are median (IQR) or % (n/N).
TAVR = transcatheter aortic valve replacement.

7.0 mm Hg, and the median effective orifice area was 1.6 cm², with no more than mild prosthetic regurgitation in all but 1 patient ([Table 4](#)).

PRIMARY ENDPOINT. At the time of the primary analysis, performed 2 years after enrollment of the last patient, estimated (Kaplan-Meier) death from any cause had occurred in 29.0% of patients assigned to TAVR and 37.1% of those assigned to OMT (HR: 0.83; 95% CI: 0.41-1.66; *P* = 0.60) ([Central Illustration](#)). Kaplan-Meier estimates of all-cause mortality were similar during the first 2 years of follow-up and diverged numerically thereafter. Estimated 2-year all-cause mortality was 21.7% in the TAVR group and 23.3% in the OMT group (HR: 0.95; 95% CI: 0.42-2.13), whereas the corresponding 5-year estimates were 44.9% and 56.6%, respectively (HR: 0.76; 95% CI: 0.43-1.35) ([Table 5](#)). Owing to the limited sample size and number of events, differences were not statistically significant, and the CIs were wide. A comparison of planned and observed primary endpoint events is shown in [Supplemental Table 7](#).

A Cox model adjusted for differences in baseline characteristics showed no difference in the primary endpoint ([Supplemental Table 8](#)).

Cardiovascular death at 2 years had occurred in 15.3% of patients assigned to TAVR and 21.1% of those assigned to OMT (HR: 0.73; 95% CI: 0.30-1.76). A listing of all deaths including potential causes is given in [Supplemental Table 9](#).

CLINICAL TRAJECTORIES. Individual patient trajectories over the entire follow-up period are shown for the TAVR group in [Figure 2](#) and for the OMT group in [Figure 3](#), depicting deaths (color-coded by adjudicated cause), heart-failure hospitalizations, permanent pacemaker implantations, strokes or transient

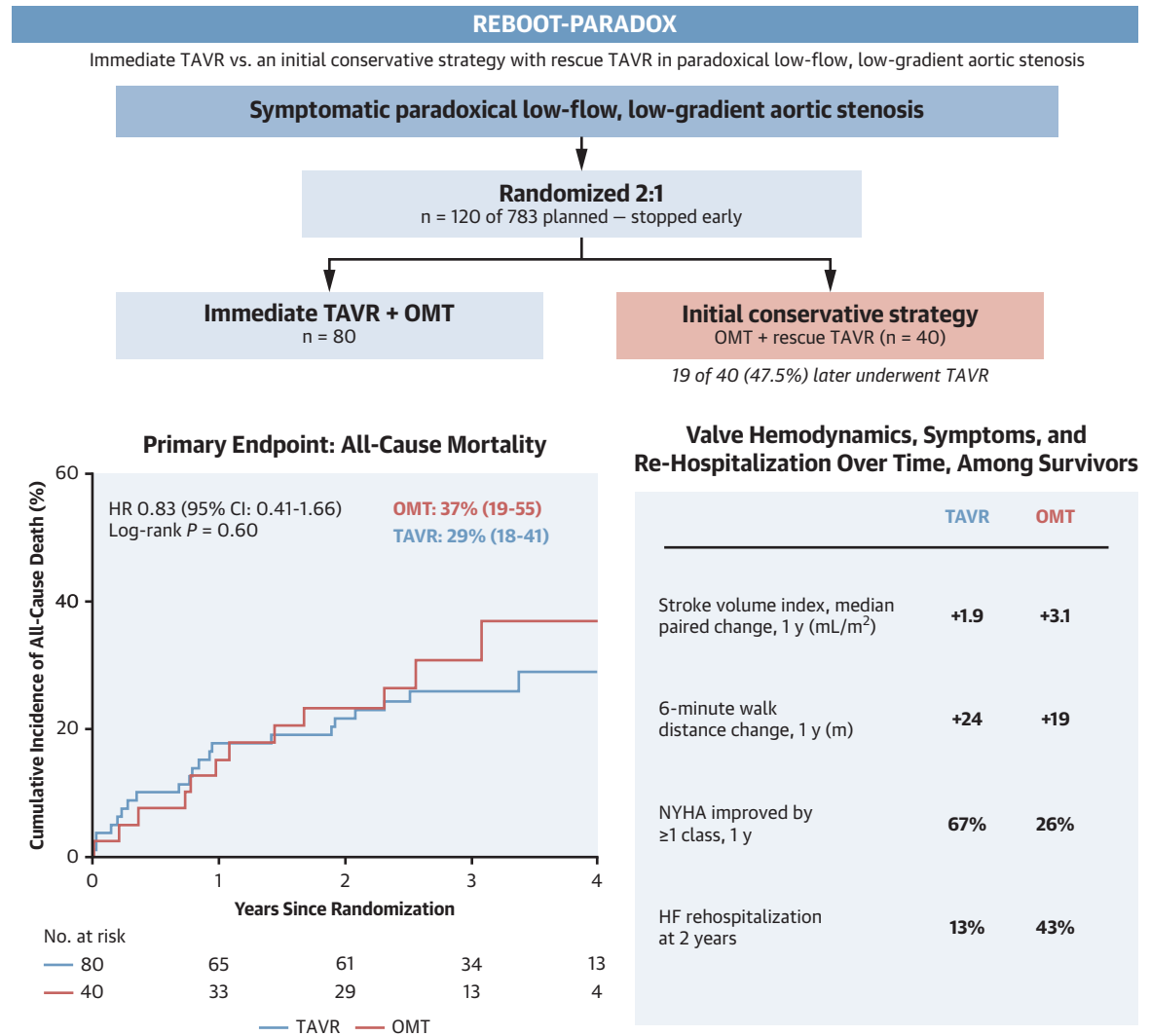
TABLE 4 Outcomes after TAVR

	TAVR-Treated (n = 79)
Technical success (VARC-3)	97.5 (77/79)
Conversion to open-heart surgery	0.0 (0/79)
Need for a second prosthesis	0.0 (0/79)
Cerebrovascular event	0.0 (0/79)
Bleeding complication	3.8 (3/79)
Vascular (access-related) complication	2.5 (2/79)
Myocardial infarction	0.0 (0/79)
Acute kidney injury	2.5 (2/79)
Need for blood-product transfusion	3.8 (3/79)
Cardiac or vascular surgery	1.3 (1/79)
Percutaneous coronary intervention	1.3 (1/79)
Pacemaker or ICD implantation	16.5 (13/79)
New or worsened conduction disturbance	34.2 (27/79)
ECG at discharge	
New AV block	7.2 (5/69)
New LBBB/RBBB	11.6 (8/69)
Echocardiography (post-TAVR)	
Prosthesis orifice area, cm ²	1.6 (1.4-2.0)
dPmean, mm Hg	7.0 (5.0-10.0)
dPmax, mm Hg	12.0 (9.0-17.0)
Regurgitation > grade 1	1.3 (1/77)

Values are % (n/N) or median (IQR).
AV = atrioventricular; ECG = electrocardiogram; ICD = implantable cardioverter defibrillator; LBBB = left bundle branch block; RBBB = right bundle branch block; TAVR = transcatheter aortic valve replacement; VARC = Valve Academic Research Consortium; other abbreviations as in [Tables 2 and 3](#).

ischemic attacks, and, in the OMT group, crossover to TAVR. Individual rationale for crossover to TAVR is presented in [Table 6](#). Over follow-up there were 32 deaths, 11 heart-failure hospitalizations, 16 pacemaker implantations, and 5 strokes in the TAVR group, and 19 deaths, 16 heart-failure hospitalizations, 4 pacemaker implantations, and 1 stroke in the OMT group. Crossover to TAVR frequently preceded subsequent events. These displays are descriptive and are not adjusted for differing exposure or timing of crossover.

SECONDARY ENDPOINTS. TAVR resulted in sustained improvement in valve hemodynamics. At 1 year, the median AVA increased from 0.8 cm² to 1.4 cm², and the median dPmean decreased from 21 mm Hg to 7 mm Hg, whereas both parameters remained largely unchanged in the OMT group ([Supplemental Figures 1A and B](#)). The stroke volume index rose modestly over time in both groups (median within-patient change at 1 year, +1.9 mL/m² in the TAVR group and +3.1 mL/m² in the OMT group), but the median remained at or below the 35 mL/m² low-flow threshold through 2 years ([Supplemental Figure 1C](#)), indicating that the low-flow state

CENTRAL ILLUSTRATION REBOOT-PARADOX: Immediate TAVR vs an Initial Conservative Strategy With Rescue TAVR in Paradoxical Low-Flow, Low-Gradient Aortic Stenoses

Mehilli J, et al. JACC. 2026;■(■):■-■.

Patients with symptomatic paradoxical low-flow, low-gradient severe aortic stenosis were randomized 2:1 to immediate transcatheter aortic valve replacement plus optimal medical therapy (TAVR, n = 80) or an initially conservative strategy with optimal medical therapy alone (OMT, n = 40); the trial enrolled 120 of 783 planned patients and was stopped early. The left panel shows the primary endpoint, the cumulative incidence of all-cause death (intention-to-treat, censored 2 years after the last patient was randomized), with the number of patients at risk below the axis. Curve labels give the cumulative incidence at the end of follow-up with 95% CIs. The groups did not differ significantly (HR: 0.83; 95% CI: 0.41-1.66; log-rank $P = 0.60$). The right panel summarizes, among survivors, hemodynamic, symptom, and rehospitalization measures by randomized group. Stroke volume index is the median within-patient paired change from baseline to 1 year; the 6-minute walk distance value is the change in the median distance at 1 year (no paired within-patient data available); NYHA functional class improvement is the proportion of assessed patients improving by at least 1 class at 1 year; heart-failure rehospitalization is the cumulative incidence of heart failure (HF) associated rehospitalizations through 2 years. Stroke volume index rose modestly in both groups and remained at or below the 35 mL/m² low-flow threshold, functional class was more favorable, and the cumulative incidence of first heart-failure rehospitalization differed between the randomized groups. Among patients initially assigned to OMT, rehospitalizations for clinical deterioration frequently led to rescue TAVR. All analyses are descriptive and not adjusted for multiplicity; follow-up measures are limited to surviving, assessed patients, and the trial was open-label, which may bias subjective measures such as NYHA functional class.

frequently persisted despite relief of the trans-valvular gradient.

New or worsened cardiac conduction disturbances occurred more frequently after TAVR than after OMT alone (48.0% vs 25.7% at 2 years; HR: 2.52; 95% CI: 1.33-4.77), and 16.7% of patients in the TAVR arm required implantation of a permanent pacemaker. Rates of stroke, periprocedural (30-day) bleeding, acute kidney injury, and other endpoints are shown in **Table 5**. Because these secondary analyses were exploratory and not adjusted for multiplicity, they should be interpreted as hypothesis-generating.

PATIENT-REPORTED OUTCOMES AND FUNCTIONAL CAPACITY. Functional status improved in both groups but to a greater extent after TAVR. Among surviving patients with an available 1-year assessment, 83% of those assigned to TAVR were in NYHA functional class I or II, compared with 59% of those assigned to OMT (**Figure 4**). In a longitudinal ordinal analysis that accounted for repeated measures, the odds of being in a lower (better) functional class favored TAVR at 1 year (OR: 0.30; 95% CI: 0.12-0.77) and attenuated by 2 years (OR: 0.53; 95% CI: 0.19-1.45); complete-case and death-inclusive displays and the full model are provided in **Supplemental Figure 2**. Health status and functional capacity showed no clinically meaningful between-group differences and remained largely unchanged over time in both groups. The EQ-5D Index changed by +0.01 in the TAVR group and −0.03 in the OMT group at 1 year (both nonsignificant), and the 6-minute walk distance and the WHO-5 Well-Being Index likewise did not differ materially between groups (**Supplemental Figures 3 to 5**).

BLOOD PRESSURE AND DIURETIC THERAPY. Systolic blood pressure was higher in the TAVR group than in the OMT group at 30 days and at 2 years, whereas diastolic pressure differed only at 2 years (**Supplemental Figure 6**). Loop-diuretic use increased over time in the OMT group (from 45% at baseline to 71% at 2 years) but not in the TAVR group (52% at baseline to 42% at 2 years) (**Supplemental Figure 7**), possibly due to a lower congestive burden after valve replacement.

The cumulative incidence of first endpoint-related rehospitalizations was 15% in the TAVR arm vs 51% in the OMT arm (HR: 0.25; 95% CI: 0.12-0.52). A similar result was seen for the incidence of rehospitalization for heart failure (**Supplemental Figure 8**).

EXPLORATORY ANALYSES. Results of the treated-as-randomized analysis, which includes only patients who received their assigned treatment during the index hospitalization (79 TAVR and 38 OMT),

TABLE 5 Clinical Outcomes

Endpoint	TAVR (n = 80)	OMT (n = 40)	HR (95% CI)	P Value
Primary endpoint				
All-cause death, 2 y after LPI	29.0% (21)	37.1% (12)	0.83 (0.41-1.66)	0.60
Mortality				
All-cause death at 1 y after randomization	17.7% (14)	15.3% (6)	1.18 (0.45-3.08)	0.73
All-cause death at 2 y after randomization	21.7% (17)	23.3% (9)	0.95 (0.42-2.13)	0.90
All-cause death at 5 y after randomization	44.9% (32)	56.6% (19)	0.76 (0.43-1.35)	0.35
Cardiovascular death at 2 y after randomization	15.3% (12)	21.1% (8)	0.73 (0.30-1.76)	0.48
Other secondary endpoints at 2 y				
MACCE	16.5% (13)	18.5% (7)	0.94 (0.38-2.32)	—
Stroke	6.4% (5)	2.7% (1)	2.50 (0.30-21.1)	—
New/worsened conduction disturbance	48.0% (38)	25.7% (10)	2.52 (1.33-4.77)	—
Combined efficacy endpoint	38% (30)	68% (26)	—	—
Rehospitalization (endpoint-related)	15% (12)	51% (19)	0.25 (0.12-0.52)	—
Rehospitalization (due to HF)	13% (10)	43% (16)	0.27 (0.12-0.59)	—
Periprocedural (30 days)				
MACCE	8.8 % (7)	2.5 % (1)	3.57 (0.44-29.1)	—
Cardiovascular death	2.5% (2)	2.5% (1)	1.00 (0.09-10.7)	—
Bleeding	11.3% (9)	2.5% (1)	4.65 (0.59-36.9)	—
Acute kidney injury	6.3% (5)	5.1% (2)	1.24 (0.24-6.32)	—

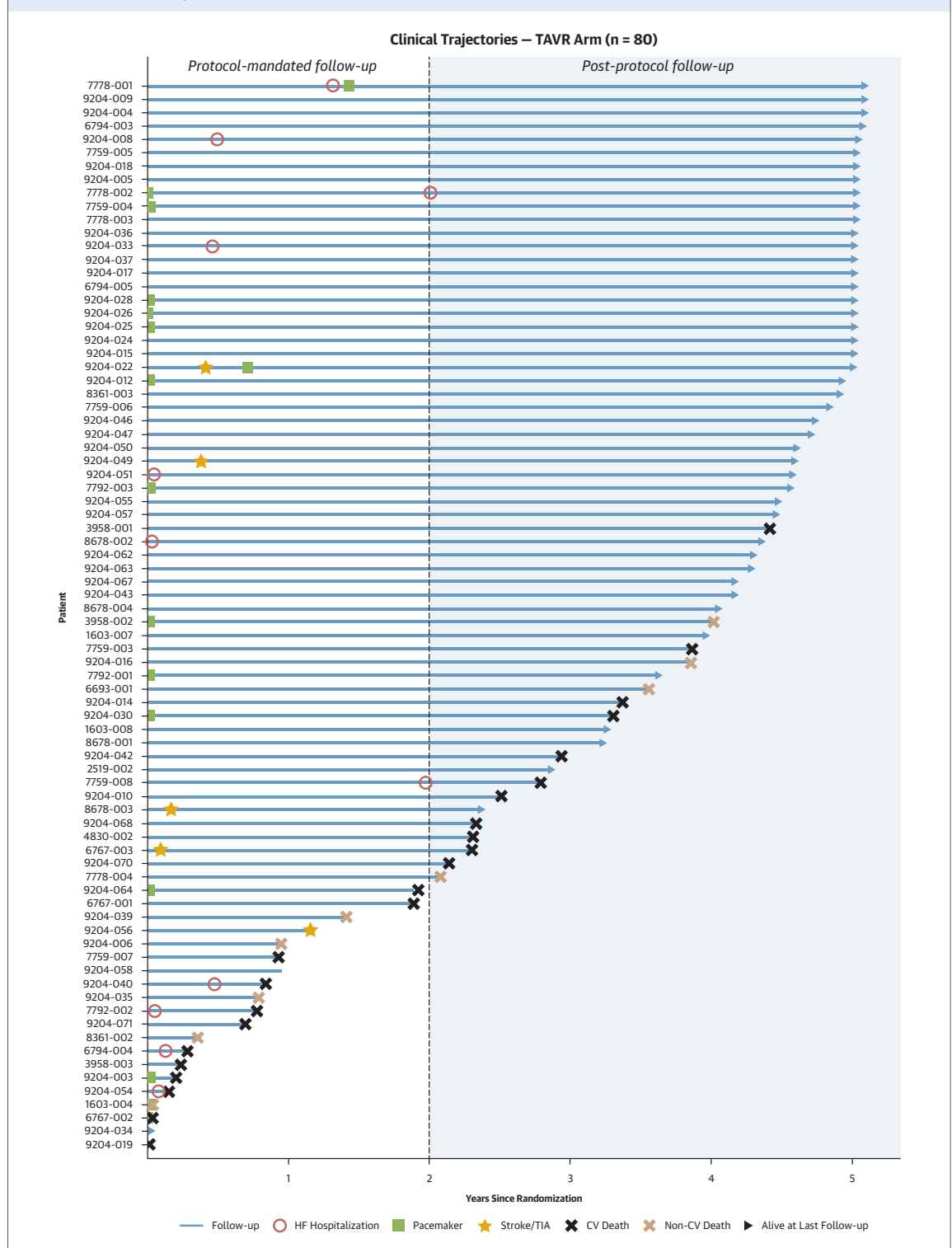
Mortality data are given as Kaplan-Meier estimates (%) and number of observed events. Numbers in parentheses are the number of patients with an event.
HF = heart failure; LPI = last patient in; MACCE = myocardial infarction, cardiac or vascular surgery, and stroke at 30 days, 1, and 2 years.

were consistent with those of the intention-to-treat analysis. At the time of the primary analysis, performed 2 years after enrollment of the last patient, all-cause mortality was 29.0% in the TAVR group and 38.9% in the OMT group (HR: 0.79; 95% CI: 0.39-1.58) (**Supplemental Figure 9**). Baseline characteristics, imaging, and medications for the treated-as-randomized population are provided in **Supplemental Tables 10 to 12**.

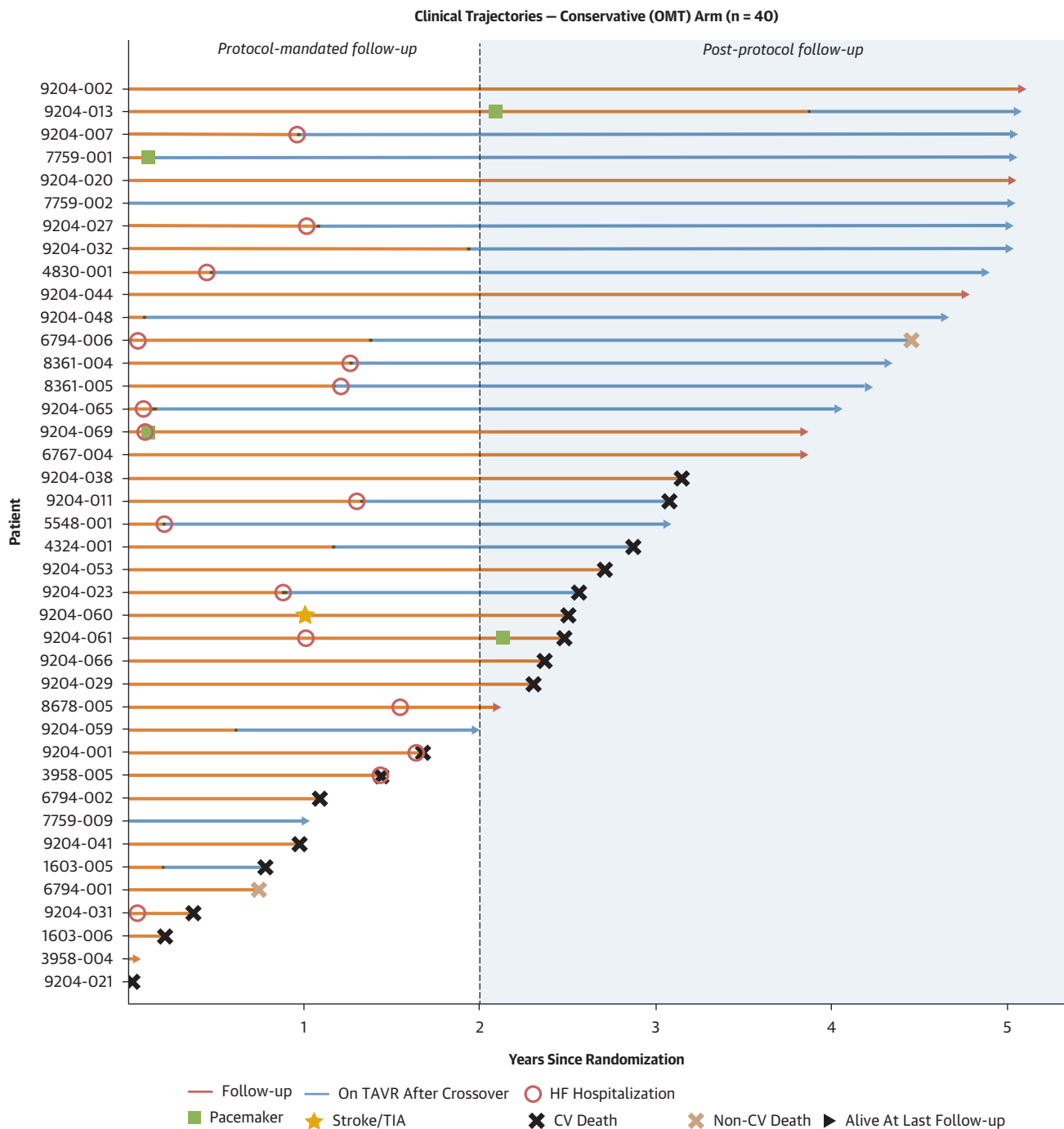
Because crossover from OMT to TAVR occurred frequently, an additional exploratory treatment-exposure analysis using a Cox proportional hazards model with TAVR included as a time-dependent covariate was performed and can be found in **Supplemental Table 13**.

DISCUSSION

The REBOOT-PARADOX trial is the first randomized trial specifically dedicated to patients with pLFLG AS. Although premature termination left the effect of TAVR on mortality unresolved, the randomized treatment allocation, extended patient-level follow-

FIGURE 2 Clinical Trajectories in the TAVR Arm

Each horizontal bar represents 1 patient randomized to transcatheter aortic valve replacement (TAVR) plus optimal medical therapy (OMT) (n = 80), ordered by follow-up duration, with time since randomization on the horizontal axis. Symbols denote heart-failure hospitalization (open red circle), permanent pacemaker implantation (green square), stroke or transient ischemic attack (orange star), and death color-coded by adjudicated cause (cardiovascular, black; noncardiovascular, magenta); an arrowhead marks patients alive at last follow-up. In keeping with Valve Academic Research Consortium 3 (VARC-3), deaths of undetermined cause are shown as cardiovascular (CV). The dashed vertical line at 2 years marks the end of protocol-mandated follow-up; beyond this point (shaded) endpoint ascertainment was event-driven. HF = heart failure; TIA = transitory ischemic attack.

FIGURE 3 Clinical Trajectories in the Conservative (OMT) Arm

Each horizontal bar represents 1 patient randomized to OMT alone (n = 40), ordered by follow-up duration. Bars are red while the patient received OMT and turn blue from the time of crossover to TAVR onward. Symbols denote heart-failure hospitalization (open red circle), permanent pacemaker implantation (green square), stroke or transient ischemic attack (orange star), and death color-coded by adjudicated cause (cardiovascular, black; noncardiovascular, magenta); an arrowhead marks patients alive at last follow-up. As in [Figure 2](#), deaths of undetermined cause are classified as cardiovascular per VARC-3. Nineteen patients crossed over from OMT to TAVR, all before the primary-endpoint cutoff. The dashed vertical line at 2 years marks the end of protocol-mandated follow-up; beyond this point (shaded) endpoint ascertainment was event-driven. Abbreviations as in [Figure 2](#).

TABLE 6 Listing of Crossovers From OMT to TAVR

Patient	Randomization-to-Crossover-Interval, Days	Reasons for Crossover ^a
7759-002	2	Treating physician's decision
7759-009	2	Treating physician's decision
7759-001	35	Repeat hospitalization due to heart failure which are associated with AS Severe progression in symptoms which is incompatible with QOL
9204-048	35	Severe progression in symptoms which is incompatible with QOL
9204-065	54	Severe progression in symptoms which is incompatible with QOL
1603-005	71	Severe progression in symptoms which is incompatible with QOL
5548-001	75	Severe progression in symptoms which is incompatible with QOL
4830-001	173	Repeat hospitalization due to heart failure which are associated with AS Severe progression in symptoms which is incompatible with QOL
9204-059	224	Severe progression in symptoms which is incompatible with QOL
9204-023	324	Repeat hospitalization due to heart failure which are associated with AS Severe progression in symptoms which is incompatible with QOL Increase of mean pressure gradient by >20 mm Hg within 6 mo
9204-007	353	Severe progression in symptoms which is incompatible with QOL
9204-027	394	Severe progression in symptoms which is incompatible with QOL
4324-001	428	Severe progression in symptoms which is incompatible with QOL Increase of mean pressure gradient by >20 mm Hg within 6 months
8361-005	443	Severe progression in symptoms which is incompatible with QOL
8361-004	463	Severe progression in symptoms which is incompatible with QOL
9204-011	484	Severe progression in symptoms which is incompatible with QOL
6794-006	505	Repeat hospitalization due to heart failure which are associated with AS
9204-032	706	Increase of mean pressure gradient by >20 mmHg within 6 mo Severe progression in symptoms which is incompatible with QOL
9204-013	1,415	Severe progression in symptoms which is incompatible with QOL

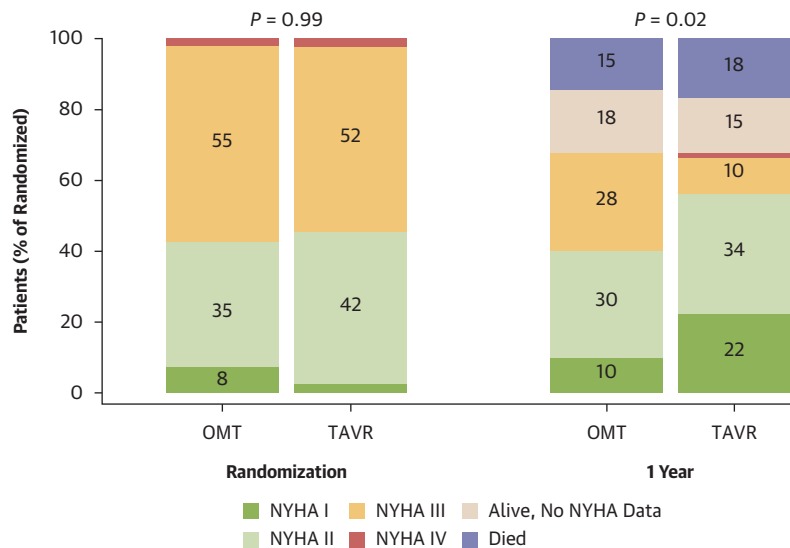
All 19 crossovers from OMT to TAVR occurred before the primary-endpoint cutoff. ^aReasons are described as in case report form.
AS = aortic stenosis; QOL = quality of life; other abbreviations as in [Table 1](#).

up, and frequent crossover provide an unusual opportunity to characterize the clinical course following immediate TAVR vs an initially conservative strategy. Nearly one-half of patients initially assigned to OMT subsequently underwent TAVR, most commonly within the first 12 to 18 months and frequently after heart-failure hospitalization, whereas others remained clinically stable for several years without intervention. Despite successful relief of valvular obstruction, heart-failure events and mortality remained frequent, and low-flow physiology persisted after TAVR. Together, these findings highlight the marked clinical heterogeneity of pLFLG AS and suggest that factors beyond the severity of valve obstruction contribute substantially to long-term outcomes.

PATIENT POPULATION. The enrolled population is representative of pLFLG AS as encountered elsewhere. Patients carried a high burden of cardiovascular disease, including atrial fibrillation (58%), arterial hypertension (88%), and coronary artery disease (58%). This profile closely matches the pLFLG

phenotype described in large single-center and registry cohorts.^{10,13} The distribution of hemodynamic parameters in our trial—median AVA 0.8 cm², mean gradient 21 mm Hg, LVEF 58%, and SVi 29.6 mL/m²—confirms that a genuinely pLFLG AS population was enrolled, rather than patients with pseudo-severe stenosis or measurement artifact. The comparability of our cohort with these prior descriptions supports the external validity of the findings.

COMPARISON TO THE EXISTING LITERATURE. The primary endpoint result must be interpreted against a body of observational data suggesting that patients with pLFLG AS derive comparatively limited benefit from AVR. Registry analyses have repeatedly shown that pLFLG stenosis carries a worse prognosis than high-gradient disease and that, among all hemodynamic subtypes, pLFLG patients gain the least symptomatic and prognostic benefit from TAVR.^{10,14} The reasons are likely intrinsic to the phenotype: the low transvalvular gradient reflects reduced flow generated by a small, stiff ventricle and extensive extra-valvular cardiac damage, so that relieving the

FIGURE 4 NYHA Functional Class Over Time

Distribution of NYHA functional class at randomization and 1 year later in patients assigned to OMT alone or TAVR plus OMT (intention-to-treat population). Bars are expressed as a percentage of all randomized patients in each group; patients who had died (purple) or who were alive but without an available NYHA assessment (peach) are displayed together with those in a known NYHA functional class (green to red). Deaths are cumulative from randomization. Abbreviations as in [Figure 2](#).

valvular obstruction addresses only 1 component of a multifactorial disease. Our finding that mortality remained high in both arms—approaching 1 in 4 at 2 years and roughly one-half at 5 years—reinforces the view that the adverse prognosis of pLFLG AS is driven substantially by comorbidity and myocardial and extra-cardiac disease that valve replacement alone may not reverse. The modest recovery in SVi observed in both treatment groups may further indicate limited myocardial recovery following TAVR in this population. Although comparable data are lacking for pLFLG AS, registry data from broader TAVR populations have shown improvement in SVi in approximately 30% of patients, without a corresponding reduction in mortality.¹⁵

Only 1 previous randomized trial has attempted to address a related question. The ROTAS (Randomised study for the Optimal Treatment of symptomatic patients with low-gradient severe Aortic valve Stenosis) trial randomized symptomatic patients with low-gradient severe AS and preserved LVEF to AVR or OMT, but was likewise stopped early for slow recruitment after enrolling 52 patients. It too failed to demonstrate a benefit of intervention on its composite endpoint of death or cardiovascular hospitalization (19% vs 27%; HR: 0.76; 95% CI:0.24-2.39).¹⁶

REBOOT-PARADOX and ROTAS thus converge on the same difficulty (recruiting adequately powered cohorts in this population is extremely challenging) and on the same directionally favorable but statistically inconclusive point estimate for intervention. An important distinction is that ROTAS enrolled a mixed low-gradient population with preserved LVEF (ie, pLFLG as well as normal-flow low-gradient AS), whereas REBOOT-PARADOX restricted eligibility to the pLFLG phenotype (ie, SVi <35 mL/m²) and required confirmation of stenosis severity by calcium scoring or prespecified likelihood criteria. This mechanistically homogeneous design is a strength of the present trial, because the heterogeneous normal-flow low-gradient AS cohort may comprise pseudo-severe AS and is otherwise known to resemble, at least in parts, high-gradient AS patients¹⁷; their post-TAVR mortality rates are lower than that of pLFLG patients.⁷

CLINICAL SURVEILLANCE AND CROSSOVER. This study compared an interventional treatment with OMT and rescue intervention. In the OMT arm, crossover rates were high and mainly driven by severe symptom progression or recurrent heart failure hospitalization. Similarly, in EARLY-TAVR (Evaluation of TAVR Compared to Surveillance for

Patients With Asymptomatic Severe Aortic Stenosis), rehospitalizations were more frequent with conservative management and largely driven by symptom progression.¹⁸ To some extent, this is a subjective endpoint and may be due to “subtraction anxiety,” which can affect patients and physicians likewise.¹⁹ Withholding a potentially beneficial therapy from a highly symptomatic patient remains challenging despite procedural risk and inconclusive evidence.

Importantly, crossover should not be interpreted as failure of the initial conservative strategy. Rather, it underscores the heterogeneity within this patient population: while a substantial proportion of patients developed clinical indications for TAVR during follow-up, others remained clinically stable for several years without intervention. These observations support an individualized management approach based on close clinical surveillance, timely intervention upon deterioration, and shared decision-making.

THERAPY OF HFpEF. The interpretation of these results is further complicated by the substantial evolution of background medical therapy since the trial was designed. pLFLG AS shares much of its pathophysiology and comorbidity profile with HFpEF, and the management of HFpEF has changed fundamentally over the enrollment and follow-up period. Sodium-glucose cotransporter-2 inhibitors and, more recently, the nonsteroidal mineralocorticoid-receptor antagonist finerenone have each demonstrated not only symptomatic but also prognostic benefit.²⁰⁻²²

Because these agents were not part of standard care when REBOOT-PARADOX began, the OMT arm of our trial does not reflect what would today be considered fully optimized medical therapy. Whether contemporary OMT might have narrowed the functional gap between the treatment arms remains to be investigated.

STUDY LIMITATIONS. This trial has important limitations. Foremost, premature termination after enrolling only 15% of the planned sample left the study substantially underpowered; the primary analysis is descriptive, CIs are wide, and no confirmatory conclusion about mortality can be drawn. Second, the high clinically indicated crossover rate diluted the contrast between the randomized groups and reduced the effective difference in treatment exposure. The treated-as-randomized and treatment-exposure analyses are exploratory and, because treatment was no longer randomly assigned after crossover, are subject to guarantee-time and selection bias. They must not be read as evidence of a

mortality benefit. Third, the open-label design may have influenced the assessment of symptom-based secondary endpoints, although mortality and adjudicated events were assessed by a blinded committee. Fourth, secondary endpoints were exploratory and not adjusted for multiple testing, so the observed differences in functional status, hemodynamics, and conduction disturbances should be regarded as hypothesis-generating. Finally, as discussed, medical therapy for the associated HFpEF phenotype advanced during the trial, so the control arm may not represent contemporary optimal care.

CONCLUSIONS

This multicenter randomized trial compared immediate TAVR with OMT and rescue TAVR in patients with pLFLG AS. Although the effect of TAVR on mortality remains unresolved, the findings suggest that an initial conservative strategy with close surveillance and timely intervention upon clinical deterioration may be appropriate for selected patients. Adequately powered randomized trials and improved risk stratification are needed to determine whether TAVR improves prognosis and to identify patients most likely to benefit from early intervention.

ACKNOWLEDGEMENTS The authors acknowledge the valuable support of the Steering Committee, the data safety monitoring board, and the event adjudication committee boards. The authors thank Janina Neubarth-Meyer, Martina Schulz, Julius Fischer, and Johannes Schurr for their support.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This study was funded by an unrestricted research grant from Boston Scientific. The funder had no role in the design or conduct of the trial, data collection, data analysis, data interpretation, or preparation of the manuscript. Dr Mehilli has received a research grant from Boston Scientific; and has received speaker honoraria from Abbott Vascular, Edwards Lifesciences, Novo Nordisk, and SMT. Dr Steffen has received speaker honoraria from Johnson&Johnson and AstraZeneca; and has received travel support from Medtronic and Shock-Wave. Dr Haum has received speaker honoraria from AstraZeneca and BMS; and has received educational grant from Biotronic. Dr Joner has received personal fees from Recor; has received grants and personal fees from Biotronic/Teleflex, Boston Scientific, and Edwards Lifesciences; has received grant support from Amgen outside the submitted work; and has received funding from the German Center for Cardiovascular Research: DZHK (FKZ 81Z0600502; FKZ 81X2600526) and from the Leducq Foundation (grant agreement number 18CVD02). Dr Adam has received honoraria from Abbott, Boston Scientific, Edwards Lifesciences, JenaValve, Medtronic, and Meril. Dr Zeymer has received personal fees from Apontis, AstraZeneca, BMS, Boehringer Ingelheim, Bayer Healthcare, Chiesi, CytoSorbents Inc, Daiichi-Sankyo, Eli Lilly, Ferrer, Merillife, Novartis, Pfizer, Sanofi, and Zoll. Dr Windecker has received research, travel, and/or educational grants to the institution from

Abbott, Abiomed, Alnylam, Amicus Therapeutics, Amgen, Anandic Medical Systems, Anteris, AstraZeneca, Bayer, B.Braun, Bioanalytica, Biotronik, Boehringer Ingelheim, Boston Scientific, Bristol Myers Squibb, Cordis Medical, CorFlow Therapeutics, CSL Behring, Daiichi-Sankyo, Edwards, Eli Lilly (Suisse), Lifesciences, Fumedica, GE Healthcare, Guerbet, IACULIS, Inari Medical, Janssen AI, Johnson & Johnson, Linde Gas, M.A. Medalliance, Medtronic, MSD Merck Sharp & Dohme, Novartis, Novo Nordisk, OM Pharma, Orchestra BioMed, Pfizer, Philips AG, Sanofi-Aventis, Servier, Siemens Healthcare, Sinomed, SMT Sahajanand Medical Technologies, TRiCares GmbH, Vascular Medical, V-Wave, and ZOLL Medical; serves as advisory board member, consultant, and/or member of the steering/executive group of trials funded by Abbott, Amgen, Anteris, Abiomed, Edwards Lifesciences, EnCarda Inc, Medtronic, Novartis, and Sinomed, with payments to the institution but no personal payments; and is a member of the steering/executive committee group of several investigator-initiated trials that receive funding by industry without impact on his personal remuneration. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Prof Julinda Mehilli, LA-Regio Kliniken, Klinik Landshut-Achdorf, Achdorfer Weg 3, Landshut 84036, Germany. E-mail: julinda.mehilli@la-regio-kliniken.de. OR Prof Steffen Massberg, Medizinische Klinik und Poliklinik I, LMU Klinikum, Marchioninistr. 15, Munich 81377, Germany. E-mail: steffen.massberg@med.uni-muenchen.de.

PERSPECTIVES

COMPETENCY IN PATIENT CARE AND PROCEDURAL

SKILLS: pLFLG AS is a heterogenous condition requiring individualized management. New conduction disturbances and permanent pacemaker implantation were more frequent after TAVR, whereas the cumulative incidence of first study-defined rehospitalization and subsequent need for rescue TAVR were higher among patients initially assigned to OMT. An initial conservative strategy with close surveillance and timely TAVR upon clinical deterioration may be appropriate in selected patients, recognizing that TAVR has no proven survival benefit, and substantial residual risk persists after intervention. This tradeoff should inform shared decision-making.

TRANSLATIONAL OUTLOOK: Adequately powered randomized trials incorporating contemporary heart-failure therapy and accounting for the high likelihood of crossover are needed to determine whether TAVR improves prognosis in pLFLG AS and to identify the clinical and imaging characteristics that distinguish patients who can safely be observed from those most likely to benefit from early intervention.

REFERENCES

- Praz F, Borger MA, Lanz J, et al. 2025 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J*. 2025;46:4635-4736.
- Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA Guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2021;77(4):450-500.
- Blankenberg S, Seiffert M, Vonthein R, et al. Transcatheter or surgical treatment of aortic-valve stenosis. *N Engl J Med*. 2024;390:1572-1583.
- Reddy RK, Howard JP, Mack MJ, et al. Transcatheter vs surgical aortic valve replacement in lower-risk patients: an updated meta-analysis of randomized controlled trials. *J Am Coll Cardiol*. 2025;85:926-940.
- Clavel M-A, Magne J, Pibarot P. Low-gradient aortic stenosis. *Eur Heart J*. 2016;37:2645-2657.
- Dayan V, Vignolo G, Magne J, Clavel MA, Mohty D, Pibarot P. Outcome and impact of aortic valve replacement in patients with preserved LVEF and low-gradient aortic stenosis. *J Am Coll Cardiol*. 2015;66:2594-2603.
- Steffen J, Reissig N, Andreae D, et al. TAVI in patients with low-flow low-gradient aortic stenosis-short-term and long-term outcomes. *Clin Res Cardiol*. 2022;111:1325-1335.
- Herrmann HC, Pibarot P, Hueter I, et al. Predictors of mortality and outcomes of therapy in low-flow severe aortic stenosis. *Circulation*. 2013;127:2316-2326.
- Ueyama H, Kuno T, Harrington M, et al. Impact of surgical and transcatheter aortic valve replacement in low-gradient aortic stenosis: a meta-analysis. *JACC Cardiovasc Interv*. 2021;14:1481-1492.
- Puls M, Beuthner BE, Topci R, et al. Patients with paradoxical low-flow, low-gradient aortic stenosis gain the least benefit from TAVI among all hemodynamic subtypes. *Clin Res Cardiol*. 2025;114:1479-1490.
- Ozkan A, Hachamovitch R, Kapadia SR, Tuzcu EM, Marwick TH. Impact of aortic valve replacement on outcome of symptomatic patients with severe aortic stenosis with low gradient and preserved left ventricular ejection fraction. *Circulation*. 2013;128:622-631.
- McDonagh TA, Metra M, Adamo M, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42:3599-3726.
- Wagener M, Reuthebuch O, Heg D, et al. Clinical outcomes in high-gradient, classical low-flow, low-gradient, and paradoxical low-flow, low-gradient aortic stenosis after transcatheter aortic valve implantation: a report from the SwissTAVI Registry. *J Am Heart Assoc*. 2023;12:e029489.
- Tribouilloy C, Rusinaru D, Marechaux S, et al. Low-gradient, low-flow severe aortic stenosis with preserved left ventricular ejection fraction: characteristics, outcome, and implications for surgery. *J Am Coll Cardiol*. 2015;65:55-66.
- Gallone G, Islas F, Gorla R, et al. Stroke volume index and transvalvular flow rate trajectories in severe aortic stenosis treated with TAVR. *Eur Heart J Cardiovasc Imaging*. 2023;24:1052-1061.
- Galli E, Le Ven F, Coisne A, et al. Randomised study for the optimal treatment of symptomatic patients with low-gradient severe aortic valve stenosis and preserved left ventricular ejection fraction (ROTAS trial). *Heart*. 2024;110:1223-1230.
- Steffen J, Andreae D, Nabauer M, et al. TAVI for patients with normal-flow low-gradient compared to high-gradient aortic stenosis. *Int J Cardiol*. 2023;371:299-304.
- Genereux P, Schwartz A, Oldemeyer JB, et al. Transcatheter aortic-valve replacement for

asymptomatic severe aortic stenosis. *N Engl J Med*. 2025;392:217–227.

19. Rajkumar CA, Nijjer SS, Cole GD, Al-Lamee R, Francis DP. 'Faith healing' and 'subtraction anxiety' in unblinded trials of procedures: lessons from DEFER and FAME-2 for end points in the ISCHEMIA trial. *Circ Cardiovasc Qual Outcomes*. 2018;11:e004665.

20. Anker SD, Butler J, Filippatos G, et al. Empagliflozin in heart failure with a preserved

ejection fraction. *N Engl J Med*. 2021;385:1451–1461.

21. Solomon SD, McMurray JJV, Claggett B, et al. Dapagliflozin in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med*. 2022;387:1089–1098.

22. Solomon SD, McMurray JJV, Vaduganathan M, et al. Finerenone in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med*. 2024;391:1475–1485.

KEY WORDS heart failure with preserved ejection fraction, low-gradient aortic stenosis, paradoxical low-flow, transcatheter aortic valve replacement

APPENDIX For supplemental tables and figures as well as a statistical analysis plan and clinical investigation plan, please see the online version of this paper.